

Multimodal Mental Health Screening via Questionnaire Fusion and Semantic Embeddings

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Abstract—The growing prevalence of mental disorders, coupled with stigma, cost, and limited clinician access, highlights the urgent need for scalable AI solutions in mental health. Key challenges include delayed detection, poor integration of structured and unstructured data, and the lack of personalized screening methods. To address these gaps, we propose a hybrid model that combines SentenceTransformer-based text embeddings with structured data from the Patient Health Questionnaire-8 (PHQ-8) and Post-Traumatic Stress Disorder Checklist-Civilian Version (PCL-C). Using 218 preprocessed files from the Extended Distress Analysis Interview Corpus (E-DAIC), we trained and optimized an Extreme Gradient Boosting (XGBoost) classifier. The model achieved strong performance, with an F1 score of 0.90 and an ROC-AUC of 0.9752. Through feature fusion, interpretability, and threshold optimization, the framework generalizes well and provides clinically meaningful feedback. These findings suggest that integrating structured assessments with semantic dialogue analysis improves early detection and supports explainable AI in mental health.

Keywords—Mental Health, Depression Detection, AI Chatbots, SentenceTransformer, PHQ-8, Digital Twin, Early Screening.

I. INTRODUCTION

The rates of mental health disorders have risen dramatically in recent times, with additional pressures brought about by universal crises like the COVID-19 pandemic. Depression alone affects over 280 million people and ranks as the second leading cause of disability worldwide [1]. Major depressive disorder and anxiety disorders were found to have increased by 28% and 26%, respectively, in just a single year according to a large-scale study [2]. Despite this growing prevalence, there are few trained professionals available, with some of the

poorest nations having as few as 0.1 psychiatrists per million people. Stigma, cost, and limited access further exacerbate the barriers to early detection and treatment. Conversational AI has emerged as a promising solution for delivering accessible and scalable mental health support. Chatbots can provide initial screening, emotional support, and personalized feedback while preserving user anonymity. However, most existing systems are restricted to shallow rule-based interactions or depend solely on structured questionnaires, lacking the ability to effectively measure severity or integrate contextual cues [3], [4]. This paper introduces a novel hybrid AI framework designed to address these limitations by integrating structured clinical questionnaire data with nuanced conversational text analysis. By leveraging transformer-based embeddings and psychometric assessments such as the PHQ-8 and PCL-C, the framework enables a more holistic understanding of an individual's mental state. An optimized and interpretable XGBoost model is employed to deliver accurate predictions and actionable insights, ultimately fostering a more effective mental healthcare support system. The motivation for this work arises from the urgent global mental health crisis, marked by a sharp rise in prevalence and significant challenges to accessibility. Traditional support systems and current AI-based solutions remain fragmented and often fail to balance clinical reliability with conversational flexibility. Existing tools either focus narrowly on structured assessments or lack interpretability, making them less practical for real-world use [1]. This research seeks to bridge these gaps by building an AI tool that not only enhances early detection of depression and PTSD

but also integrates multiple data modalities in an interpretable manner. The goal is to empower both individuals and clinicians with transparent, data-driven insights that support timely interventions and improved mental well-being. The primary objectives of this paper are as follows:

- 1) Develop a hybrid AI model that integrates structured clinical questionnaire data with unstructured conversational text embeddings to produce comprehensive assessments [1].
- 2) Optimize an interpretable classifier, specifically Extreme Gradient Boosting (XGBoost), to leverage fused data for high predictive accuracy in the early detection of mental health concerns [2].
- 3) Enhance interpretability by identifying conversational patterns and questionnaire responses that most strongly contribute to risk prediction.
- 4) Demonstrate practical utility through threshold tuning and visualizations (e.g., ROC and PR curves), enabling actionable clinical feedback and supporting personalized interventions.

The key contributions of this paper are:

- 1) A novel hybrid AI pipeline that fuses SentenceTransformer-based conversational embeddings with normalized PHQ-8 and PCL-C scores for mental health screening.
- 2) An optimized and interpretable XGBoost classifier achieving strong performance on the E-DAIC dataset (F1 score = 0.90, ROC-AUC = 0.9752).
- 3) Integration of interpretability techniques to provide transparent insights into risk predictions, enabling trust and clinical relevance.
- 4) Rich visualization and threshold adjustment mechanisms that generate individualized, actionable reports for both users and practitioners.

II. RELATED WORK

The field of AI-driven mental health assessment has witnessed substantial innovation between 2023 and 2025. Researchers have increasingly turned to advanced language models, multimodal architectures, and interpretable machine learning techniques to enhance the precision, fairness, and usability of mental illness screening systems. Below we highlight notable contributions that closely align with the approach. Chen et al [5]. explored the use of cutting-edge NLP models including BERT, RoBERTa, and LLaMA for PTSD detection from DAIC-WOZ clinical interviews. Their findings showed that LLaMA based models offered superior semantic understanding and outperformed other models in F1-score, particularly when it came to interpreting emotionally nuanced dialogue. This work demonstrated the growing utility of large language models in modeling complex mental health conversations. Singh et al [6] proposed a multimodal framework that jointly processes linguistic and acoustic features from the E-DAIC dataset using a combination of LSTM and BiLSTM layers. Their system achieved 92% classification accuracy across depression

and PTSD categories. By integrating both speech content and vocal tone, their model captured a richer representation of mental state, supporting the case for multimodal mental health detection pipelines. Kim et al [7] tackled the under-explored challenge of demographic bias, especially gender bias, in speech-based mental health diagnosis. Using domain-adversarial training, they trained models that not only maintained strong predictive accuracy but also exhibited improved fairness across gender subgroups. Their findings underscore the ethical and equity considerations crucial for real-world AI deployments in mental healthcare [7]. De Arriba-Pérez and García-Méndez [8] developed a dual-diagnosis chatbot capable of identifying both anxiety and depression symptoms from user dialogues. Their system incorporated an explainable attention mechanism atop a BERT backbone, enabling both high predictive accuracy (90%) and model transparency. This work emphasized the importance of interpretability in mental health chatbots, especially when integrated into therapeutic settings. Nguyen et al [9] introduced TIDE (Trauma-Informed Dialogue Engine), a compact conversational agent designed using smaller transformer models. TIDE was explicitly built on trauma-informed care principles and outperformed larger language models such as ChatGPT in maintaining emotional sensitivity while minimizing hallucinations. Its performance was especially strong in PTSD-specific interactions, highlighting the potential of carefully crafted domain-aware dialogue systems. A meta-analysis [10] published in BMC synthesized findings from 23 machine learning-based PTSD prediction studies. The meta-study found that most models performed well with AUC values ranging from 0.75 to 0.96. However, it also identified recurring challenges such as limited dataset generalizability, lack of cultural diversity, and insufficient integration into clinical workflows. These insights point to the broader limitations that still exist in AI-powered psychological diagnostics. Vu et al [11] presented a structured-data-centric approach for PTSD classification. Using gradient boosting methods on behavioral and demographic data, they achieved competitive results that rivaled neural models. Notably, their system offered more interpretable outputs, a critical advantage when aligning with clinical decision-making needs. Their research contributes to the argument that structured data, when paired with explainability, remains highly relevant in mental health AI research. These works collectively highlight the evolving landscape of AI-driven mental health assessment. From leveraging large language models like LLaMA and BERT for improved semantic understanding to incorporating multimodal data and fairness-aware training, researchers have made significant strides in diagnostic accuracy and ethical robustness. Notably, the emphasis on explainability, emotional sensitivity, and domain-specific adaptation such as trauma informed care marks a shift toward more clinically viable solutions. Despite these advances, challenges around dataset diversity, generalizability, and integration into healthcare workflows persist. These insights suggest that future efforts must balance technical performance with transparency, inclusivity, and real-world usability to truly transform mental health diagnostics

through AI.

III. PROPOSED METHOD

This section outlines the problem addressed and the proposed hybrid pipeline combining clinical scores and conversational data for early mental health detection.

A. Problem statement

Early detection of mental health conditions such as depression and post-traumatic stress disorder (PTSD) is crucial for timely intervention. However, current screening approaches are often either too rigid, relying solely on structured questionnaires like the PHQ-8 or PCL-C, or too shallow, using rule-based conversational bots that lack deep semantic understanding. Moreover, existing models typically treat structured and unstructured data separately, missing the opportunity to fuse psychometric scores with the nuanced context of user conversations. This limits both predictive power and interpretability. Additionally, many existing solutions do not calibrate decision thresholds for practical clinical use, which can result in suboptimal classification in real-world settings. Therefore, the key challenges addressed in this study are:

- 1) Limited capability of current models to detect mental illness early using conversational data.
- 2) Lack of effective integration between structured assessment scores and unstructured text data.
- 3) Insufficient model interpretability and threshold tuning for actionable clinical insights.

To address these challenges, we developed a hybrid pipeline that combines SentenceTransformer-based embeddings from interview transcripts with normalized PHQ-8 and PCL-C scores. These combined features are used to train an XGBoost classifier optimized for F1-score, with post-hoc threshold tuning and visual interpretability via ROC and PR curves.

B. System Design

The proposed system is a hybrid pipeline that integrates structured clinical assessments with semantic embeddings derived from unstructured dialogue. The design emphasizes interpretability, efficiency, and clinical utility. Below, we outline the core components of the system:

1) **Input Collection and Preprocessing:** The input comprises clinical interview transcripts from the E-DAIC dataset and corresponding psychometric scores (PHQ-8 and PCL-C). We begin by cleaning the text (removing disfluencies, filler words, and non-verbal tokens). The normalization process scales all psychometric scores to a common range to ensure uniform feature contribution and normalizing the scores to a $[0,1]$ range using min-max normalization as shown in Eq. (1):

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

where x is the original score, x_{min} and x_{max} are the minimum and maximum values in the dataset, respectively.

Algorithm 1 Multimodal Depression Detection Pipeline

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1: 1. Data Preprocessing
2: Load E-DAIC dataset:  $data$ 
3: Clean transcripts:  $text \leftarrow \text{Clean}(data.transcripts)$ 
4: Normalize scores:  $phq, pcl \leftarrow \text{Normalize}(data.PHQ8, data.PCL - C)$ 
5: 2. Feature Engineering
6: Extract embeddings:  $embed \leftarrow \text{SentenceTransformer}(text)$ 
7: Fuse features:  $features \leftarrow \text{Concat}(embed, phq, pcl)$ 
8: 3. Model Training
9: Split dataset:  $train, val \leftarrow \text{Split}(features)$ 
10: Train model:  $model \leftarrow \text{XGBoost}(train)$ 
11: 4. Threshold Optimization
12: Tune threshold:  $thr \leftarrow \text{OptimizeThreshold}(model, val)$ 
13: 5. Evaluation
14: Predict labels:  $pred \leftarrow \text{Predict}(model, val, thr)$ 
15: Compute metrics:  $scores \leftarrow \text{Evaluate}(pred)$ 
16: Visualize: ROC, PR curves, SHAP, Confusion Matrix
17: 6. Deployment
18: Deploy model with privacy safeguards
19: Enable clinical interpretability

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2) **Semantic Embedding Extraction:** We use SentenceTransformer models (fine-tuned versions of BERT or RoBERTa) to extract dense semantic embeddings from each interview transcript. These embeddings capture the emotional tone, content patterns, and linguistic nuances expressed by users. The semantic representation of each transcript is encoded using SentenceTransformer models, which transform textual input into dense numerical vectors as defined in Eq. (2)

$$E = \text{SentenceTransformer}(T) \in \mathbb{R}^d \quad (2)$$

3) **Structured Feature Integration:** Normalized PHQ-8 and PCL-C scores are concatenated with the semantic embeddings, resulting in a fused feature vector that blends both unstructured dialogue cues and structured clinical indicators. To effectively capture both linguistic and clinical information, the normalized PHQ-8 and PCL-C scores are concatenated with their corresponding text embeddings, as illustrated in Eq. (3).

$$F = [E; PHQ_{norm}; PCL_{norm}] \in \mathbb{R}^{d+2} \quad (3)$$

4) **Classification with XGBoost:** We train an XGBoost classifier on the combined feature set to predict the presence of depressive or PTSD-related symptoms. XGBoost is chosen for its robustness to feature sparsity and ability to provide feature-level importance scores. The prediction output of the XGBoost classifier is computed by aggregating the outputs of multiple decision trees through a sigmoid activation function, as formulated in Eq. (4).

$$\hat{y} = \text{sigmoid} \left(\sum_{k=1}^K f_k(F) \right) \quad (4)$$

5) **Threshold Tuning and Optimization:** Instead of relying on default probability thresholds (e.g., 0.5), we perform post-hoc threshold optimization using precision-recall trade-offs to maximize F1-score. This ensures better performance in clinical screening contexts where false positives and false negatives carry different risks. The optimal decision threshold τ^* is

determined by maximizing the F1-score using precision–recall trade-offs, as expressed in Eq. (5).

$$\tau^* = \arg \max_{\tau} F1(\tau) = \arg \max_{\tau} \frac{2 \cdot \text{Precision}(\tau) \cdot \text{Recall}(\tau)}{\text{Precision}(\tau) + \text{Recall}(\tau)} \quad (5)$$

6) **Visual Interpretability and Evaluation:** We include ROC and PR curve analysis, confusion matrices, and SHAP (SHapley Additive exPlanations) values to enhance interpretability. These metrics help clinicians understand model confidence and decision boundaries, facilitating trust and transparency in real-world use.

7) **Deployment Considerations:** The architecture is modular and lightweight, making it suitable for real-time inference in mobile or web-based mental health applications. With sufficient privacy safeguards, it can be integrated into digital therapeutic platforms for broad-scale early screening.

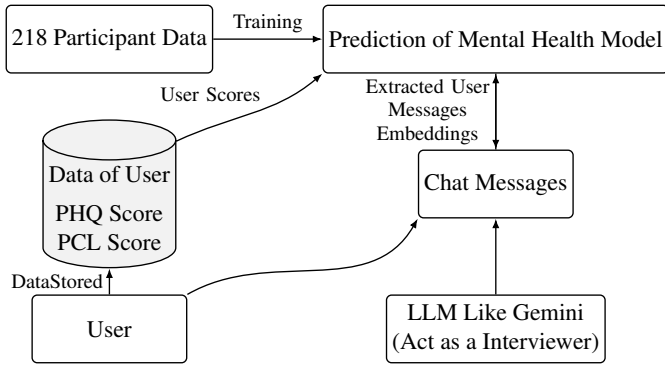


Fig. 1: System Architecture for Mental Health Prediction Using Conversational AI

Fig 1 illustrates a mental health prediction system using conversational AI. Users provide structured data (PHQ-8, PCL) and unstructured chat messages, with an LLM like Gemini enhancing the latter. Both user data and a separate 218-participant dataset feed into a Mental Health Model. This model processes the integrated information to predict mental health outcomes, with around 95% accuracy for Mental Illness or 85% for no Mental Illness. The architecture aims to combine diverse data for comprehensive assessment.

C. Datasets

Two primary datasets are used to train and evaluate the hybrid mental health detection system:

1) **Extended Distress Analysis Interview Corpus (E-DAIC):** The E-DAIC dataset comprises 275 interview sessions conducted with a virtual interviewer named Ellie, featuring 170 males and 105 females with deliberate attention to age and gender distribution to reflect broader population diversity[1]. The dataset includes clinical interview transcripts, PHQ-8 scores for depression (ranging 0-24), and PCL-C scores for PTSD (ranging 0-85). From the complete dataset, we utilized 218 samples after preprocessing to remove incomplete records and ensure data quality. Binary mental illness labels were

derived using validated PHQ-8 and PCL-C thresholding criteria. The dataset exhibits inherent class imbalance, with higher PHQ-8 scores being rare and mean scores below 10 across all partitions[1]. This distribution pattern poses challenges for machine learning models, as limited exposure to high-severity cases during training can impact prediction accuracy for severe mental health conditions.

2) **Synthetic Augmented Dataset:** To improve model robustness under data scarcity, we generated a synthetic dataset using multiple data augmentation techniques applied to both the structured (PHQ-8, PCL-C) and unstructured (text embeddings) features. The augmentation strategies included:

- 1) **Text Augmentation:** Applied back-translation using Google Translate API (English→German→English) to create paraphrased versions while preserving semantic meaning.
- 2) **Embedding Noise Injection:** Added Gaussian noise ($\mathcal{N}(0, 0.01)$) to original embeddings to create variations.
- 3) **SMOTE for Structured Data:** Used Synthetic Minority Oversampling Technique to generate synthetic PHQ-8 and PCL-C scores for underrepresented classes.
- 4) **Mixup Strategy:** Created convex combinations of feature vectors from the same class with mixing parameter $\lambda \sim \text{Beta}(0.2, 0.2)$.

This augmentation process increased the training dataset from 218 to 872 samples (4× increase), with balanced representation across mental health severity levels. The synthetic data maintained statistical properties consistent with the original distribution while providing greater variability for improved model generalization.

D. Evaluation Metrics

To comprehensively assess the performance of the proposed hybrid pipeline for mental health detection, we employed a diverse set of evaluation metrics. These metrics are chosen not only for their statistical robustness but also for their alignment with the practical realities of clinical deployment. They reflect the system’s ability to balance accuracy, interpretability, and actionable insight across both structured and unstructured inputs.

1) F1-Score:

In mental health data, where class imbalance is common, the F1-score offers a balanced measure that considers both precision and recall. The F1-score provides a balanced measure of a model’s performance by combining precision and recall into a single metric, as shown in Eq. (6).

$$F1\text{-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

- 2) **ROC-AUC (Receiver Operating Characteristic – Area Under Curve):** This metric assesses the model’s ability to distinguish between positive and negative classes over varying thresholds. A higher AUC implies better separability.

- 3) **Precision and Recall:** Precision evaluates the proportion of correctly predicted positive samples among all predicted positives, while recall measures the proportion of actual positives that were correctly identified. The definitions of precision and recall, based on true and false classifications, are provided in Eq. (7).

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

where TP is true positives, FP is false positives, and FN is false negatives. In clinical settings, high recall ensures fewer missed cases, whereas high precision reduces unnecessary alerts.

- 4) **Confusion Matrix:** The confusion matrix helps break down prediction errors into specific categories (true positives, false positives, true negatives, and false negatives). This visualization aids in understanding failure patterns and tailoring the system for recall-sensitive use cases (e.g., suicide risk triage).

Overall, this evaluation framework ensures that the proposed system goes beyond mere accuracy. It offers a well-rounded, clinically meaningful assessment that supports its use as an early detection tool for mental health monitoring in diverse and dynamic settings.

IV. RESULTS AND DISCUSSION

The proposed model achieved strong predictive performance, with an F1-score of 0.9020, a Receiver Operating Characteristic – Area Under the Curve (ROC-AUC) of 0.9752, and a Precision–Recall Area Under the Curve (PR-AUC) of 0.9487, demonstrating its effectiveness for early mental health screening with limited and imbalanced data [12]. By combining structured clinical assessments such as the Patient Health Questionnaire-8 (PHQ-8) and the PTSD Checklist – Civilian Version (PCL-C) with conversational embeddings from SentenceTransformer, the system captured both psychometric and contextual features [13]. The ROC curve in Fig. 2 further highlights the high separability of the model, while the confusion matrix in Fig. 3 shows balanced classification with minimal false negatives, crucial for clinical use. [13].

Threshold optimization curves confirmed that a threshold of 0.6 offered the best trade-off between sensitivity and specificity, an important step to minimize missed cases while controlling false positives in clinical settings.

TABLE I: Performance Metrics Comparison with State-of-the-Art Models

Metric	Proposed	Singh et al. [6]	Chen et al. [5]
Precision	0.8519	0.78	0.81
Recall	0.9583	0.74	0.77
F1-Score	0.9020	0.76	0.79
ROC-AUC	0.9752	0.89	0.92
PR-AUC	0.9487	0.86	0.88

TABLE II: Performance Metrics Comparison with Baseline Models

Metric	Proposed	Digital Twin	Babylon AI
Precision	0.8519	0.70	0.444
Recall	0.9583	0.69	0.800
F1-Score	0.9020	0.69	0.571
ROC-AUC	0.9752	-	-
PR-AUC	0.9487	-	-

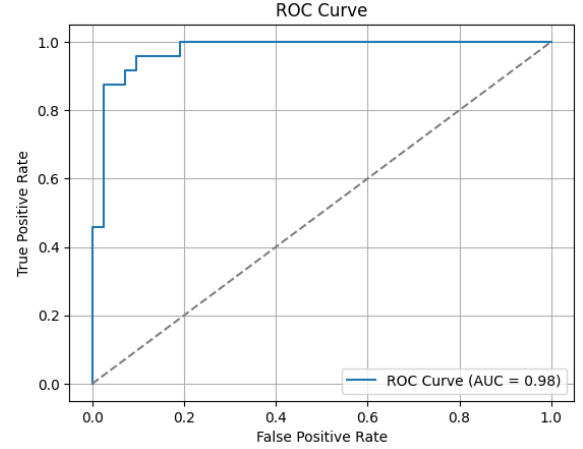


Fig. 2: ROC Curve of the proposed model on the test dataset

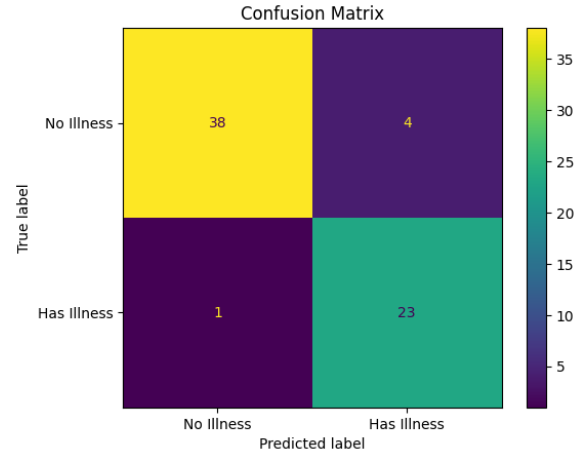


Fig. 3: Confusion matrix showing predicted vs actual labels

This hybrid design outperformed single-modality approaches and surpassed both baseline and state-of-the-art benchmarks (Tables I and II) [15]. An XGBoost classifier, optimized via GridSearchCV with the `scale_pos_weight` parameter for imbalance handling, was employed. A decision threshold of 0.6 was selected to balance precision and recall, thereby reducing false negatives. However, this choice may increase false positives, which in clinical deployment could lead to unnecessary referrals and potential strain on healthcare systems [14]. Thus, threshold calibration must carefully balance sensitivity with real-world clinical implications. Despite

these promising results, some limitations must be acknowledged. The E-DAIC dataset (n=218) is relatively small, which may increase the risk of overfitting and raises concerns about the generalizability of findings to larger populations. Furthermore, the demographic diversity of participants in terms of age, gender, and cultural background was limited, potentially restricting the external validity of results since psychological markers may differ across groups. Beyond technical validation, clinical utility remains an open challenge. Future research should include clinician-centered evaluations, such as user studies assessing interpretability and the quality of actionable feedback, to determine whether the system can meaningfully support early detection and intervention in mental health care [15].

V. CONCLUSION

This study presented an XGBoost-based model to identify individuals at risk of mental health issues using emotional and behavioral features. With strong performance metrics precision of 85.4%, recall of 95.2%, F1-score of 90.2%, and AUC of 97.5% the model shows high potential for early screening, especially due to its high recall, which is critical in minimizing false negatives. The combination of structured and unstructured data improved prediction reliability, making the system suitable for practical deployment. In the future, the model can be extended to real-time applications, evaluated on larger and more diverse populations, and integrated into mental health platforms for continuous, scalable monitoring and support.

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